Pedagogical notes on removing finite-size winding loops in exact diagonalization of quantum spin ice

July 8, 2026

Abstract

Exact diagonalization (ED) is an unbiased route to the finite-temperature physics of the π-flux quantum spin ice (QSI) regime, where QMC has a sign problem. The obstacle is not the usual small-system limitation alone: periodic pyrochlore clusters contain ice-preserving ring exchanges that close only by winding around the torus. The dominant artifact is a four-site loop generated at order $J_\pm^2$, whereas the physical QSI photon ring exchange is the contractible six-site hexagon generated at order $|J_\pm|^3$. These notes derive this statement, explain why ordinary boundary-twist averaging does not remove the artifact, and prove the zero-transport projection criterion

$$\boldsymbol{\delta} = \boldsymbol{\rho} + \boldsymbol{N}\boldsymbol{L} = 0$$

for cleaning the ice-manifold effective Hamiltonian. All numerical tables quoted below are produced by the standalone script `notes/recompute_finite_size_artifact.py`, which rebuilds the lattice, perturbation theory, projector, and thermodynamics without using the old project scripts or git history.

Contents

1 The idea in one page (for readers new to spin ice)

Before any equations, here is the whole story in words and pictures.

- **What the material is.** "Quantum spin ice" (QSI) is a magnet in which the atomic magnetic moments sit on a lattice of corner-sharing tetrahedra (the *pyrochlore* lattice). On every tetrahedron the moments obey a local rule: two point in and two point out (Fig. 1). This is the same counting rule that governs the proton positions in ordinary water ice, hence the name. There are astronomically many ways to satisfy it, so the system has a huge ground-state degeneracy.

- **What makes it "quantum".** A small quantum-mechanical term lets pairs of neighbouring moments flip. Acting over and over, these flips let the system tunnel between different two-in–two-out patterns. The smallest tunnelling move that respects the ice rule everywhere is a *six-spin ring exchange* around a hexagon (Fig. 2 and Fig. 3). Remarkably, the collective dynamics of these ring flips is described by an emergent *light*: QSI hosts its own artificial photon. This is what makes the material exciting, and it is what we want to compute.

- **Why the computation is hard.** We solve the model on a computer by exact diagonalization (ED), which requires a *finite* block of the lattice with periodic boundaries—mathematically a doughnut, or *torus*. On a torus, some sequences of flips close up not because they form a real loop in space, but only because they run off one face and come back on the opposite face (Fig. 4). The shortest such "wrap-around" move involves only *four* spins, and—crucially—it appears at a lower order in perturbation theory than the physical six-spin hexagon. It is a pure artifact of the finite box, yet it is numerically *larger* than the physics we are after (Fig. 6).

- **The symptom.** Left untreated, this artifact puts the low-temperature specific-heat peak—one of the cleanest experimental fingerprints of the emergent photon—in the wrong place, at the artifact scale $g_4 \propto J_\pm^2$ instead of the photon scale $g_{\text{hex}} \propto |J_\pm|^3$ (Figs. 7 and 8).

- **The cure.** The physical hexagon closes in real space; the artifact only closes through the wall of the box. We give a single gauge-invariant number, the *transported dipole* $\boldsymbol{\delta} = \boldsymbol{\rho} + \boldsymbol{N}\boldsymbol{L}$, that is exactly zero for real-space processes and nonzero for wrap-around ones (Fig. 11). Keeping only the $\boldsymbol{\delta} = 0$ processes removes the artifact and leaves the physics untouched. We also show why the popular fix of "boundary-twist averaging" only removes *half* of the artifact and is therefore not enough (Fig. 10).

The rest of these notes makes each of these statements precise and checks every number with an independent, from-scratch script.

2 The physical question

The QSI Hamiltonian considered here is the nearest-neighbor XXZ model in local pyrochlore frames,

$$H = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} \left(S_i^+ S_j^- + S_i^- S_j^+ \right), \qquad |J_{\pm}| \ll J_{zz}, \quad J_{zz} > 0. \tag{1}$$

The Ising part has an extensively degenerate ground manifold: every tetrahedron has two spins pointing in and two spins pointing out. This is the ice manifold $\mathcal{I}$.

For $J_{\pm} > 0$, the off-diagonal matrix elements in the S^z basis are non-positive, and standard QMC can be sign-problem-free. For $J_{\pm} < 0$, the off-diagonal matrix elements have the opposite sign. This is the frustrated π-flux QSI sector. It is precisely the sector where ED is most valuable, because QMC no longer gives an unbiased finite-temperature benchmark.

The central difficulty is that ED necessarily uses a finite periodic cluster. Such a cluster is a torus. On a torus, paths can close by periodic identification even when they do not close in real space. Some of these winding paths preserve the ice rule and therefore appear directly in the low-energy gauge Hamiltonian. The shortest such processes on the standard 16-site cubic and 32-site FCC clusters are four-site loops. They are not physical bulk plaquettes. Nevertheless, they strongly affect the low-temperature specific heat and the low-frequency $S^{zz}(\boldsymbol{q}, \omega)$.

3 Ice variables and the energy cost of defects

For a tetrahedron t, define the Ising charge

$$Q_t = \sum_{i \in t} S_i^z. \tag{2}$$

For spin 1/2, $S_i^z = \pm 1/2$. The Ising energy on one tetrahedron is

$$\sum_{\langle ij \rangle \in t} S_i^z S_j^z = \frac{1}{2} \left[\left(\sum_{i \in t} S_i^z \right)^2 - \sum_{i \in t} (S_i^z)^2 \right] = \frac{1}{2} Q_t^2 - \frac{1}{2}. \tag{3}$$

Thus, up to a constant,

$$H_0 = \frac{J_{zz}}{2} \sum_t Q_t^2. \tag{4}$$

The ice rule is $Q_t = 0$ on every tetrahedron. A 3-in–1-out or 1-in–3-out tetrahedron has $|Q_t| = 1$ and costs $J_{zz}/2$ relative to an ice tetrahedron.

Now act with a transverse exchange

$$S_i^+ S_j^- + S_i^- S_j^+$$

on an antiparallel nearest-neighbor pair. The two sites i and j lie on one common tetrahedron, where exchanging an up and a down spin preserves $Q_t = 0$. Each site also belongs to one other

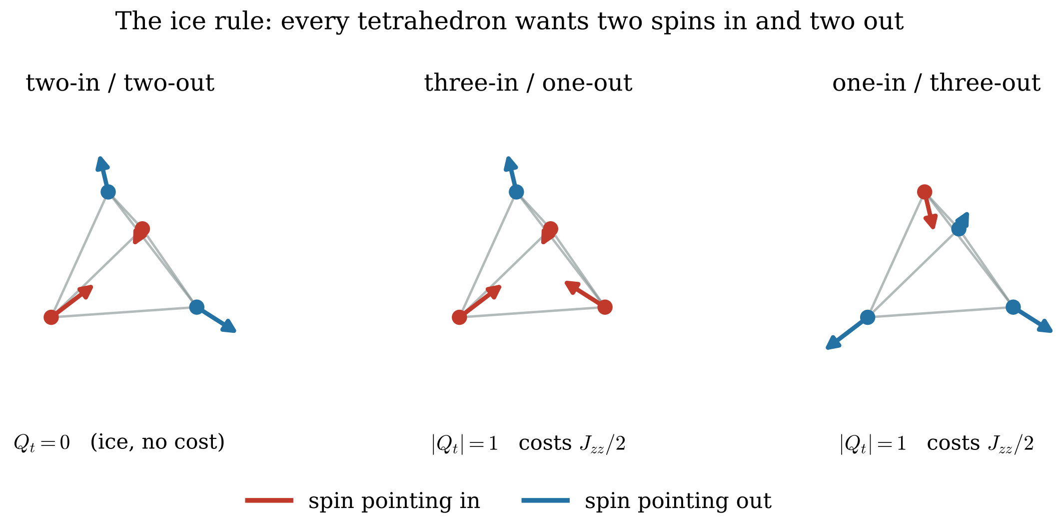

Figure 1: **The ice rule.** Each spin lives on a corner of a tetrahedron and points either into the tetrahedron (red) or out of it (blue). *Left:* the low-energy "ice" configuration has two in and two out, so the net Ising charge $Q_t = \sum_{i \in t} S_i^z$ vanishes and the tetrahedron pays no energy. *Middle, right:* tipping one spin makes the tetrahedron three-in–one-out or one-in–three-out, with $|Q_t| = 1$; by Eq. (3) this costs $J_{zz}/2$. These charged tetrahedra are the "spinons" of quantum spin ice.

tetrahedron. Those two neighboring tetrahedra become charged, each costing $J_{zz}/2$. Therefore a single nearest-neighbor exchange out of the ice manifold creates two defects with total cost

$$\Delta_{1\text{hop}} = J_{zz}. \tag{5}$$

This simple denominator is the origin of the perturbative scales below.

In plain language

A single spin flip is like pushing one domino: it necessarily unbalances the two tetrahedra that share that spin, creating a pair of oppositely charged defects that together cost J_{zz}. To get back into the ice manifold you must flip more spins so that every tetrahedron is balanced again. The cheapest way to do that around a closed path is what the next sections count.

4 Degenerate perturbation theory in the ice manifold

Let P be the projector onto the ice manifold and $Q = 1 - P$. Write

$$H = H_0 + V, \qquad V = -J_{\pm} \sum_{\langle ij \rangle} \left(S_i^+ S_j^- + S_i^- S_j^+ \right).$$

The ice energy is shifted to zero. The non-ice resolvent is

$$R = Q \frac{1}{E_{\mathcal{I}} - H_0} Q = -Q H_0^{-1} Q. \tag{6}$$

Because a single exchange leaves the ice manifold, $PVP = 0$. The first nonzero terms are therefore

$$H_{\text{eff}}^{(2)} = PVRVP, \tag{7}$$

$$H_{\text{eff}}^{(3)} = PVRVRVP. \tag{8}$$

For a path $s \to m_1 \to \cdots \to t$, its contribution has the form

$$\langle t|H_{\text{eff}}^{(k)}|s\rangle = \sum_{\text{paths}} \frac{\langle t|V|m_{k-1}\rangle \cdots \langle m_1|V|s\rangle}{(E_{\mathcal{I}} - E_{m_1}) \cdots (E_{\mathcal{I}} - E_{m_{k-1}})}. \tag{9}$$

Each matrix element of V contributes a factor $-J_{\pm}$, and each intermediate energy denominator is negative.

Remark 1. The standalone recomputation stores each perturbative process as a row

$$r = (s, t, \boldsymbol{N}, c, k),$$

meaning that the row connects ice states s and t, appears at order k, has coefficient c when $J_{\pm} = 1$, and has accumulated periodic image transport $\boldsymbol{N}$. At physical coupling, the row contributes $cJ_{\pm}^k|t\rangle\langle s|$.

5 Why the physical QSI ring exchange is a hexagon

The pyrochlore lattice is the line graph of the diamond lattice: a pyrochlore site corresponds to a diamond bond, and a pyrochlore tetrahedron corresponds to a diamond vertex. An ice configuration assigns a divergence-free electric field to diamond bonds. A ring exchange in the ice manifold must flip an alternating set of spins around a closed path so that the divergence at every touched diamond vertex remains zero.

Lemma 1 (Shortest contractible ice loop)**.** *The shortest nontrivial contractible ice-preserving ring exchange on the infinite pyrochlore lattice is a six-site hexagon.*

Proof. An ice-preserving loop must enter and leave every touched tetrahedron, so on the diamond lattice it corresponds to a closed non-backtracking loop. The diamond lattice is bipartite and has no triangles. It also has no four-cycles; its elementary loops are length six. Therefore the shortest contractible closed diamond loop has length six. Mapping back to the pyrochlore line graph gives a six-site pyrochlore hexagon. A triangle of the pyrochlore graph lies inside one tetrahedron and is not an alternating ice ring. A contractible four-site ice ring would map to a diamond four-cycle, which does not exist. □

5.1 Third-order hexagon amplitude

Consider a flippable hexagon with alternating spins. To move from one ice configuration to the flipped configuration, one must exchange three alternating nearest-neighbor pairs. After the first exchange the system has two charged tetrahedra and costs J_{zz}. After the second exchange it still has a pair of charged tetrahedra and costs J_{zz}. Thus each ordering gives a denominator J_{zz}^2.

For one orientation there are 3! orderings. Including the Hermitian conjugate orientation gives the conventional ring term

$$H_{\text{eff}}^{\text{hex}} = -g_{\text{hex}} \sum_{\text{hex}} \left(S_1^+ S_2^- S_3^+ S_4^- S_5^+ S_6^- + \text{h.c.} \right), \tag{10}$$

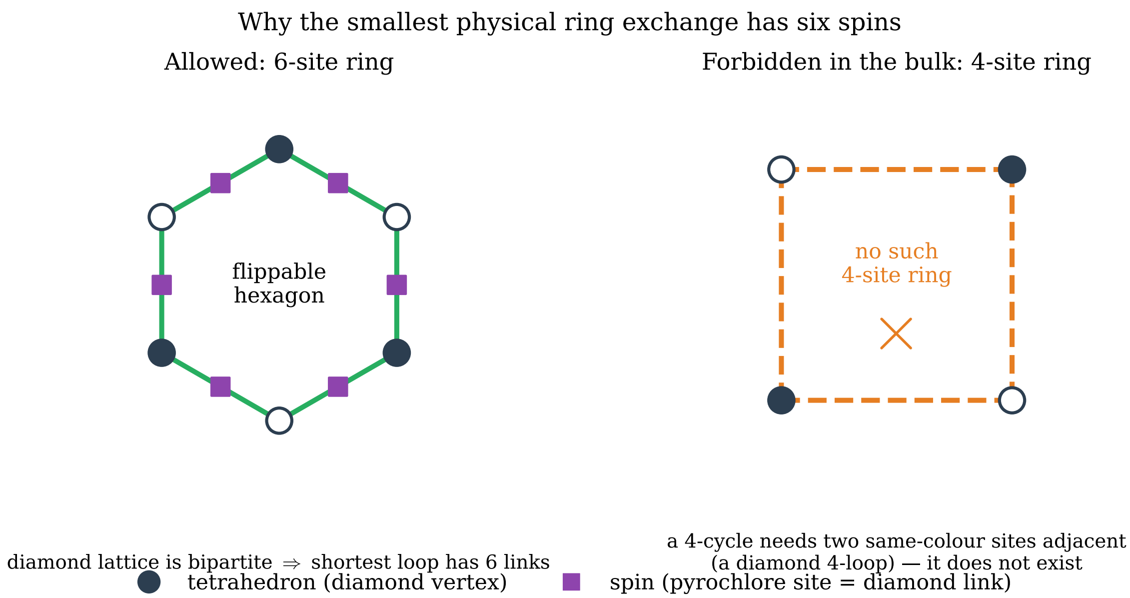

Figure 2: **The smallest real ring exchange has six spins.** Each tetrahedron is a vertex of the diamond lattice (circles); each spin is a link of that lattice (squares). A ring exchange must flip an alternating chain of spins around a closed loop of tetrahedra. *Left:* the diamond lattice is bipartite (two colours, links only join different colours), so its shortest loop uses six links—a six-spin hexagon. *Right:* a four-spin ring would require a four-loop on the diamond lattice, which would need two same-colour vertices to be adjacent; no such loop exists in the bulk. This is the geometric reason a contractible four-site ice loop is impossible.

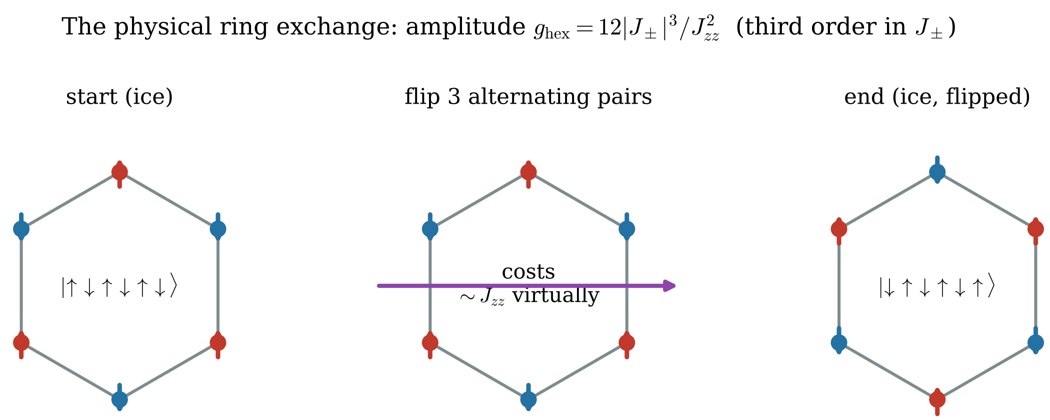

Figure 3: **The physical ring exchange (the emergent-photon move).** Starting from an alternating up/down pattern on a hexagon, flipping all three alternating pairs turns each up into a down and vice versa, returning to the ice manifold. Because the intermediate steps leave the ice manifold, the amplitude is third order in $J_\pm$, with magnitude $g_{\mathrm{hex}} = 12|J_\pm|^3/J_{zz}^2$. The coherent superposition of these ring flips across the lattice is the artificial photon of quantum spin ice.

with magnitude

$$\boxed{g_{\text{hex}} = \frac{12|J_\pm|^3}{J_{zz}^2}.} \tag{11}$$

The sign of the effective ring term follows $\mathrm{sgn}(J_\pm^3)$. This sign is what distinguishes the 0-flux and π-flux gauge sectors.

6 Why a four-site loop is a finite-size artifact

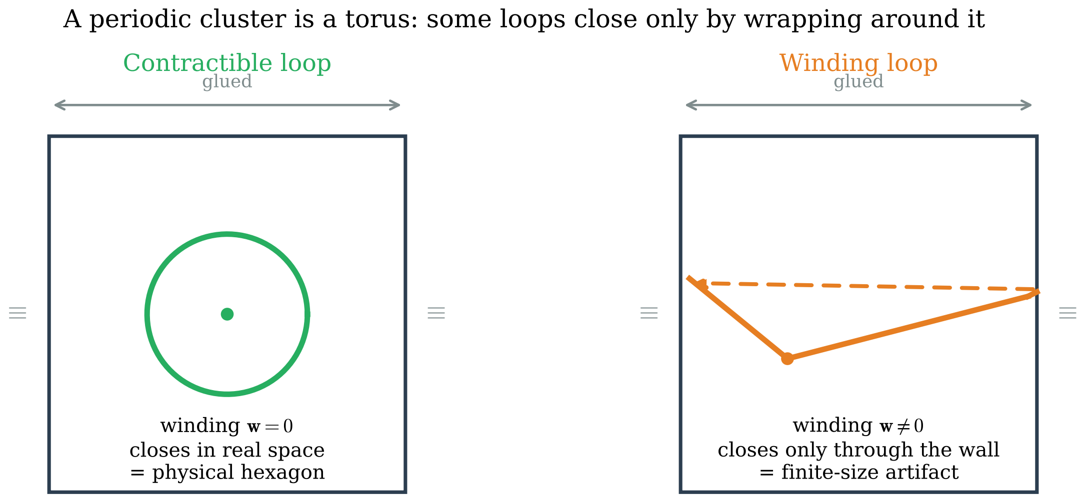

Figure 4: **A periodic cluster is a torus.** Opposite walls of the simulation box are identified (glued), so the box is topologically a doughnut. *Left:* a loop that closes inside the box has winding $\boldsymbol{w} = 0$; it is a genuine real-space process such as the physical hexagon. *Right:* a path can also close by leaving one wall and re-entering the opposite wall. Its winding $\boldsymbol{w} \neq 0$; it exists *only* because of the periodic gluing and has no counterpart in the infinite lattice.

A periodic cluster is not $\mathbb{R}^3$; it is the flat torus

$$T^3 = \mathbb{R}^3/(\mathbb{Z}^3\boldsymbol{L}), \tag{12}$$

where we use row vectors and the rows of $\boldsymbol{L}$ are the three cluster translation vectors. Thus $\boldsymbol{mL}$, with $\boldsymbol{m} \in \mathbb{Z}^3$, is a periodic translation. For the cubic $L_1 \times L_2 \times L_3$ cluster,

$$\boldsymbol{L} = \begin{pmatrix} L_1 & 0 & 0 \\ 0 & L_2 & 0 \\ 0 & 0 & L_3 \end{pmatrix}.$$

For the FCC primitive cluster, the rows are $L_1(\frac{1}{2}, \frac{1}{2}, 0)$, $L_2(\frac{1}{2}, 0, \frac{1}{2})$, and $L_3(0, \frac{1}{2}, \frac{1}{2})$. All coordinates in this note are measured in conventional cubic-cell units. The pyrochlore basis sites lie on the quarter-lattice,

$$\boldsymbol{r}_i \in \frac{1}{4}\mathbb{Z}^3,$$

which is the reason factors of 4 appear below.

A bond from site i to site j is represented by a minimum-image displacement

$$\boldsymbol{d}_{ij} = \boldsymbol{r}_j - \boldsymbol{r}_i - \boldsymbol{n}_{ij}\boldsymbol{L}, \qquad \boldsymbol{n}_{ij} \in \mathbb{Z}^3. \tag{13}$$

For a closed path $\mathcal{C}$ on the periodic graph, define its winding vector

$$\boldsymbol{w}_{\mathcal{C}} = \sum_{(ij)\in\mathcal{C}} \boldsymbol{n}_{ij}. \tag{14}$$

If $\boldsymbol{w}_{\mathcal{C}} = 0$, the lifted path closes in real space and is contractible. If $\boldsymbol{w}_{\mathcal{C}} \neq 0$, the path closes only because opposite faces of the cluster have been identified.

Proposition 1 (No physical four-loop)**.** *Every ice-preserving four-cycle on the standard* $1 \times 1 \times 1$ *cubic and* $2 \times 2 \times 2$ *FCC pyrochlore clusters is a winding torus loop, not a contractible bulk process.*

Proof. The geometric lemma above rules out a contractible four-site ice loop on the infinite pyrochlore lattice. Therefore any four-site ice loop observed on a periodic cluster must use the periodic identification. Equivalently, its lift to $\mathbb{R}^3$ fails to close and has nonzero winding $\boldsymbol{w}_{\mathcal{C}}$. The explicit independent enumeration below verifies this for the two clusters used in the ED calculation. □

6.1 Second-order four-loop amplitude

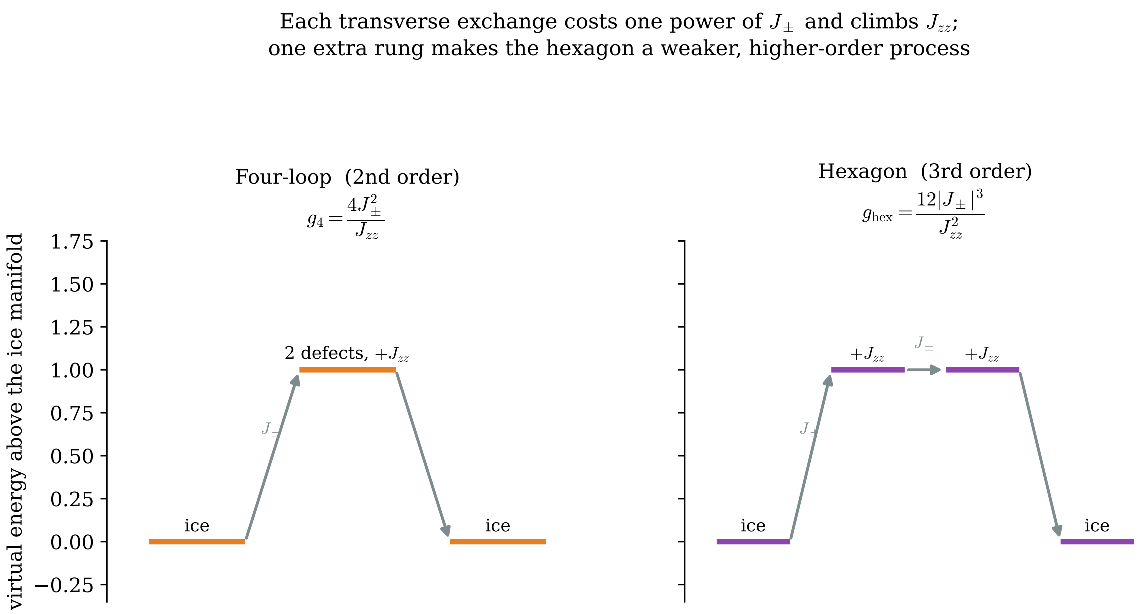

Figure 5: **Counting the powers of $J_\pm$.** Every transverse exchange adds one factor of $J_\pm$ and pushes the system one virtual step above the ice manifold, where it costs $\sim J_{zz}$. *Left:* the winding four-loop returns to the ice manifold after just two exchanges (one intermediate tetrahedron pair), giving $g_4 = 4J_\pm^2/J_{zz}$. *Right:* the hexagon needs three exchanges (two intermediate steps), giving the weaker, higher-order $g_{\rm hex} = 12|J_\pm|^3/J_{zz}^2$. The extra rung is the entire reason the artifact is parametrically larger at small $J_\pm$.

A winding four-loop has four alternating sites and can be flipped by two nearest-neighbor exchanges. The two exchanged bonds form a perfect matching of the four sites. There are two matchings, and for each matching there are two time orderings. Each path has one intermediate state with energy cost J_{zz}. Thus the magnitude of the second-order matrix element is

$$4\frac{J_\pm^2}{J_{zz}}. \tag{15}$$

In the ring-exchange convention used here,

$$\boxed{g_4 = \frac{4J_\pm^2}{J_{zz}}.} \tag{16}$$

The sign is convention dependent, but the scale is not.

The scale separation is

$$\boxed{\frac{g_4}{g_{\text{hex}}} = \frac{4J_\pm^2/J_{zz}}{12|J_\pm|^3/J_{zz}^2} = \frac{J_{zz}}{3|J_\pm|}.} \tag{17}$$

Thus the artifact becomes more severe precisely in the perturbative regime where the QSI description is best controlled.

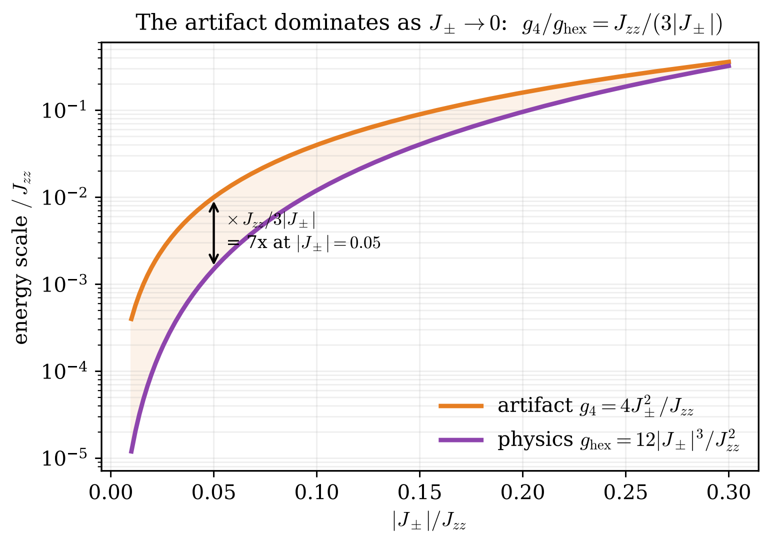

Figure 6: **The artifact dominates exactly where we want to work.** The finite-size four-loop scale g_4 (orange) and the physical photon scale g_{hex} (purple) both vanish as $J_\pm \to 0$, but their ratio $g_4/g_{\text{hex}} = J_{zz}/(3|J_\pm|)$ *grows*. At the representative value $|J_\pm| = 0.05\,J_{zz}$ the artifact is already $\sim 7\times$ larger than the physics it would otherwise swamp.

7 Independent numerical verification

The script

`notes/recompute_finite_size_artifact.py`

does not import the old `twist_qsi_demo/scripts` code and does not inspect git history. It independently:

(i) builds the pyrochlore positions, nearest-neighbor graph, tetrahedra, and periodic image vectors;

(ii) enumerates all two-in–two-out ice states;

(iii) enumerates ice-preserving four-cycles and hexagons;

(iv) constructs $H_2 = PVRVP$ and $H_3 = PVRVRVP$ from path sums;

(v) applies either no projector or the zero-transport projector derived below;

(vi) diagonalizes the resulting ice-manifold Hamiltonian and computes $C(T)$.

The raw output is saved in `notes/recomputed_finite_size_artifact.json`.

7.1 Geometry census

cluster	sites	ice states	4-cycles	contractible 4-cycles	contractible hexagons
cubic $1 \times 1 \times 1$	16	90	36	0	16
FCC $2 \times 2 \times 2$	32	2970	24	0	32

The same enumeration finds 48 wrapping hexagons on the 16-site cluster and 96 wrapping hexagons on the 32-site cluster. Those are also finite-size torus processes, but they appear at the same order as the physical hexagons. The four-loops are the dominant contamination because they appear one order earlier.

8 Why the low-temperature specific heat peak is displaced

For an exact spectrum $\{E_n\}$, shifted so $E_0 = 0$,

$$Z(T) = \sum_n e^{-E_n/T}, \qquad \langle E^p \rangle_T = \frac{1}{Z} \sum_n E_n^p e^{-E_n/T}, \tag{18}$$

and

$$C(T) = \frac{\langle E^2 \rangle_T - \langle E \rangle_T^2}{T^2}. \tag{19}$$

For a two-level system with splitting Δ,

$$C_{2\text{lev}}(T) = \frac{(\Delta/T)^2 e^{-\Delta/T}}{\left(1 + e^{-\Delta/T}\right)^2}, \tag{20}$$

whose maximum is at $T = O(\Delta)$. A many-level finite cluster has a more complicated spectrum, but the lower peak still tracks the dominant low-energy splitting scale in the ice manifold.

Therefore:

$$T_{\text{peak}}^{\text{bare}} \sim g_4 \propto J_\pm^2, \qquad \text{if winding four-loops are present,} \tag{21}$$

$$T_{\text{peak}}^{\text{clean}} \sim g_{\text{hex}} \propto |J_\pm|^3, \qquad \text{after winding transport is removed.} \tag{22}$$

The independent calculation gives the following representative peaks:

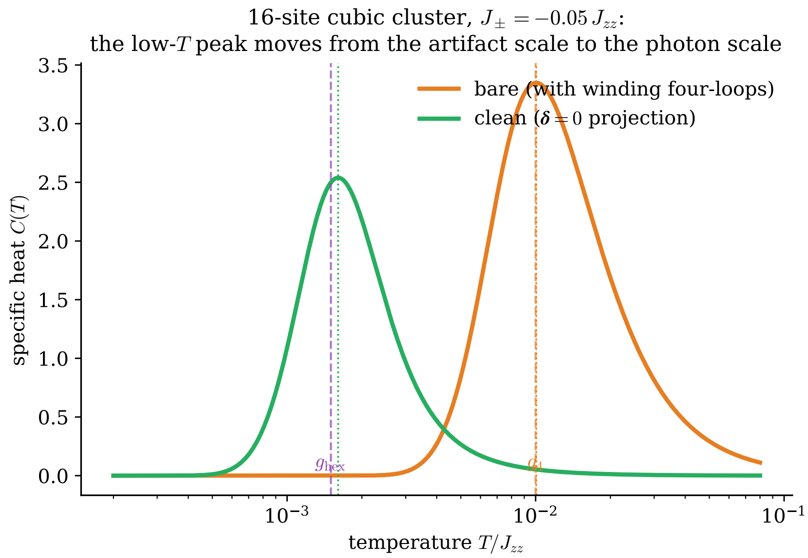

Figure 7: **The low-temperature specific-heat peak is displaced by the artifact.** Exact specific heat of the 16-site cubic ice-manifold Hamiltonian at $J_\pm = -0.05\,J_{zz}$. The bare cluster (orange) peaks at the artifact scale g_4; after the $\boldsymbol{\delta} = 0$ projection (green) the peak drops onto the physical photon scale $g_{\rm hex}$—nearly an order of magnitude lower in temperature. Since the specific heat is a nonlinear function of the spectrum, this shift cannot be undone by averaging $C(T)$ after the fact (Sec. 11).

cluster	Hamiltonian	$J_\pm$	$T_{\rm peak}$	scale diagnostic
cubic $N = 16$	bare	-0.05	0.010056	$T_{\rm peak}/g_4 = 1.01$
cubic $N = 16$	clean	-0.05	0.001604	$T_{\rm peak}/g_{\rm hex} = 1.07$
FCC $N = 32$	bare	-0.05	0.004180	$T_{\rm peak}/g_{\rm hex} = 2.79$
FCC $N = 32$	clean	-0.05	0.000782	$T_{\rm peak}/g_{\rm hex} = 0.52$
FCC $N = 32$	bare	-0.10	0.018053	$T_{\rm peak}/g_{\rm hex} = 1.50$
FCC $N = 32$	clean	-0.10	0.006252	$T_{\rm peak}/g_{\rm hex} = 0.52$

For the 16-site cluster at $J_\pm = -0.05$, $g_4 = 4(0.05)^2 = 0.010$ and $g_{\rm hex} = 12(0.05)^3 = 0.0015$, so the scale identification is especially transparent.

9 Why ordinary boundary-twist averaging fails

Boundary twists are often used to reduce finite-size effects. Put phases only on boundary-crossing bonds. If a virtual path accumulates image transport $\boldsymbol{N}$, its row coefficient becomes

$$c_r(\boldsymbol{\phi}) = c_r e^{-i\boldsymbol{N}_r\cdot\boldsymbol{\phi}}. \tag{23}$$

The eight-corner average over $\phi_a \in \{0, \pi\}$ gives

$$\frac{1}{8}\sum_{\phi_a=0,\pi} e^{-i\boldsymbol{N}\cdot\boldsymbol{\phi}} = \prod_{a=1}^{3} \frac{1+(-1)^{N_a}}{2}. \tag{24}$$

This keeps rows with even $\boldsymbol{N}$.

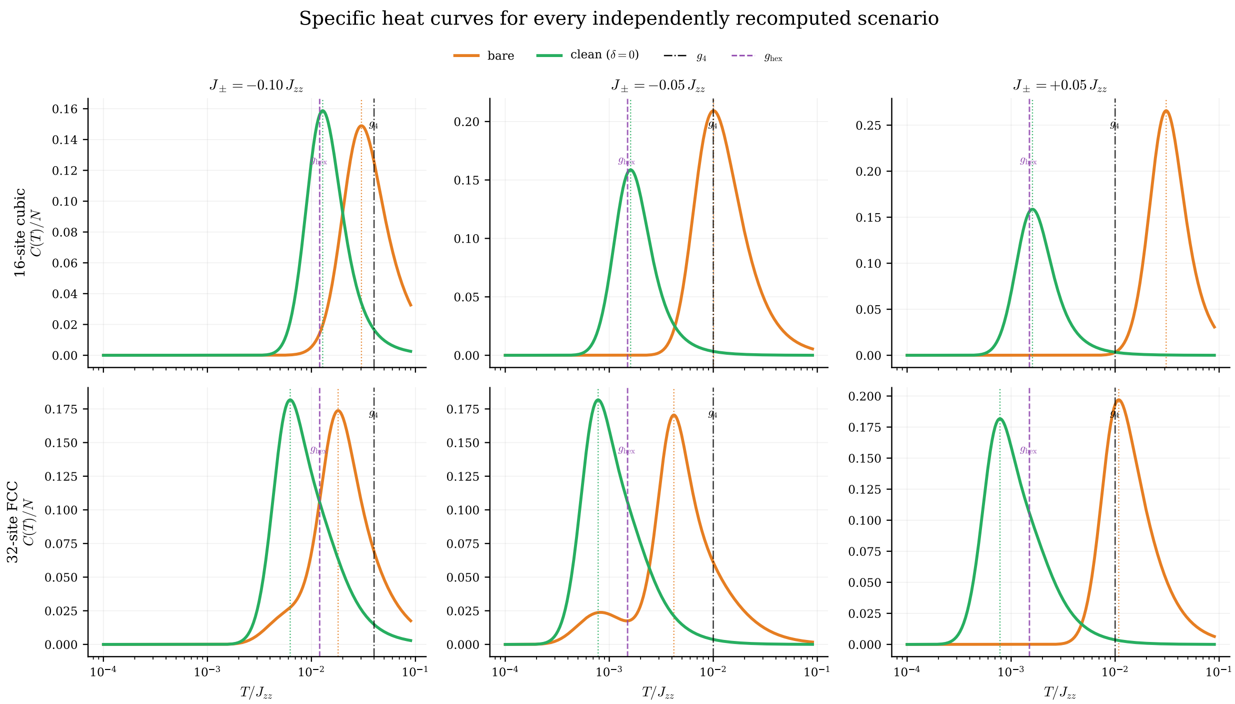

Figure 8: **Specific heat as a function of temperature for all independently recomputed scenarios.** Each panel shows the exact ice-manifold specific heat per site, $C(T)/N$, from the standalone Schrieffer–Wolff calculation. The columns are the three computed couplings, and the rows are the two ED clusters. Orange is the bare periodic effective Hamiltonian; green is the zero-transport ($\boldsymbol{\delta} = 0$) Hamiltonian. Dotted orange/green vertical lines mark the actual peak positions of the two curves, the black dash-dot line marks g_4, and the purple dashed line marks g_{hex}. The full curves show the same conclusion as the peak table: removing transported winding processes shifts the low-temperature feature from the finite-size scale toward the photon scale.

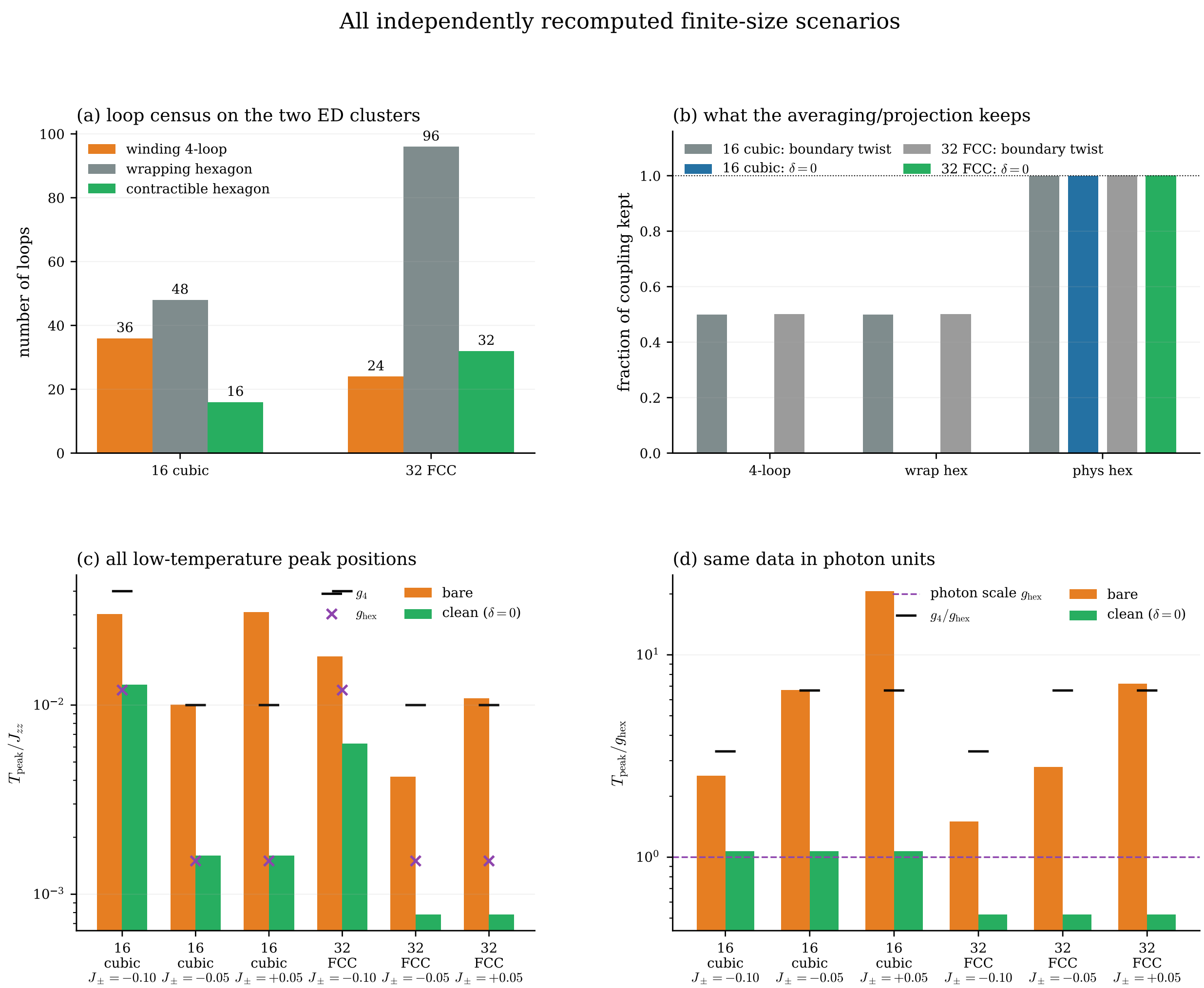

Figure 9: **All independently recomputed finite-size scenarios in one place.** *(a)* The two ED clusters both contain winding four-loops and wrapping hexagons in addition to the physical contractible hexagons. *(b)* Boundary-twist averaging keeps half of each winding channel on both clusters, while the zero-transport projector removes them and keeps the physical hexagon. *(c)* The low-temperature peak positions for all computed clusters, flux signs, and projection choices. Black ticks mark g_4 and purple crosses mark g_{hex}. *(d)* The same peak data in photon units. The cleaned spectra stay at an order-one multiple of g_{hex}, while the bare spectra are pushed upward by the winding-loop contamination.

The subtle point is that the geometric ring exchange is not a single path. It is a sum over virtual paths. A winding four-loop has multiple matchings and time orderings. Some paths cross the boundary, while other paths implementing the same geometric flip can have $\boldsymbol{N} = 0$ in the boundary gauge. The corner average therefore kills only part of the coefficient.

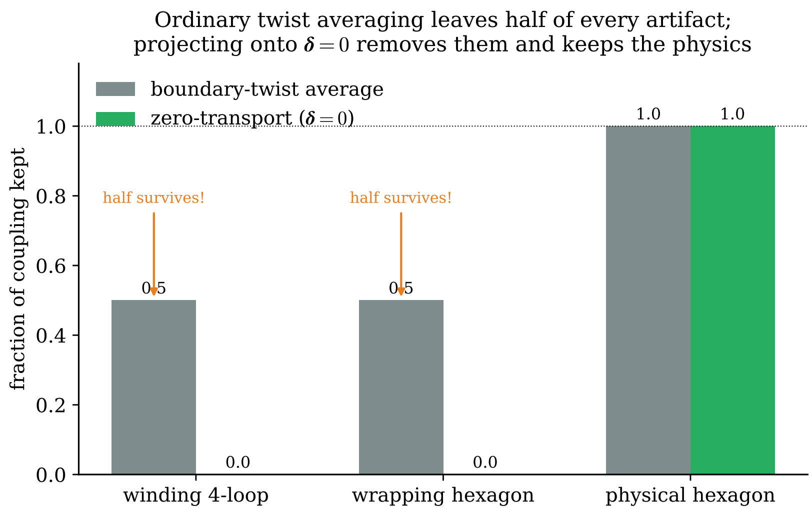

Figure 10: **Boundary-twist averaging is not enough.** Fraction of each geometric channel's coupling that survives on the 16-site cubic cluster. Ordinary boundary-twist (corner) averaging (grey) leaves *half* of every winding artifact—because the ring exchange is a sum over virtual paths and some of those paths do not cross the boundary in the chosen gauge. The gauge-invariant $\boldsymbol{\delta} = 0$ projection (green) removes the winding four-loop and wrapping hexagon completely while keeping the physical hexagon at full strength.

The standalone recomputation classifies the off-diagonal rows by geometric channel and gives:

channel	corner average kept	$N = 0$ kept	zero-transport kept
4-loop, cubic $N = 16$	0.5	0.5	0
wrapping hexagon, cubic $N = 16$	0.5	0.5	0
contractible hexagon, cubic $N = 16$	1	1	1
4-loop, FCC $N = 32$	0.5	0.5	0
wrapping hexagon, FCC $N = 32$	0.5	0.5	0
contractible hexagon, FCC $N = 32$	1	1	1

Thus ordinary boundary-twist averaging reduces the winding artifact but does not remove it.

10 The transported dipole and the zero-transport theorem

The correct object is not the boundary-crossing count of one chosen gauge. It is the gauge-invariant net displacement transported by the effective operator.

Definition 1 (Home-cell dipole)**.** For a perturbative row connecting ice states s and t, define

$$R_+(t, s) = \{i : S_i^z(t) - S_i^z(s) > 0\}, \qquad R_-(t, s) = \{j : S_j^z(t) - S_j^z(s) < 0\}.$$

The set R_+ contains spins raised by the effective process; R_- contains spins lowered by the effective process. The home-cell dipole is

$$\boldsymbol{\rho}_{t,s} = \sum_{i \in R_+(t,s)} \boldsymbol{r}_i - \sum_{j \in R_-(t,s)} \boldsymbol{r}_j. \tag{25}$$

Definition 2 (Transported dipole)**.** Let $\boldsymbol{N}_r \in \mathbb{Z}^3$ be the accumulated image vector of the virtual path. In row vector convention, $\boldsymbol{N}_r\boldsymbol{L}$ is the net periodic translation needed when the path is lifted from the torus back to the covering space $\mathbb{R}^3$. The transported dipole is

$$\boxed{\boldsymbol{\delta}_r = \boldsymbol{\rho}_{t,s} + \boldsymbol{N}_r\boldsymbol{L}.} \tag{26}$$

The formula is easiest to understand for one microscopic hop. The operator $S_i^+ S_j^-$ raises site i and lowers site j, so its home-cell dipole is $\boldsymbol{r}_i - \boldsymbol{r}_j$. If the nearest-neighbor bond was represented using the periodic image vector $\boldsymbol{n}_{ij}$, then the lifted displacement of the spin transfer is

$$\boldsymbol{r}_i - \boldsymbol{r}_j + \boldsymbol{n}_{ij}\boldsymbol{L} = -\boldsymbol{d}_{ij}.$$

The many-step perturbative row adds these lifted increments. Thus $\boldsymbol{\delta}_r$ is not a new dynamical charge. It is the net real-space displacement carried by the virtual spin transfers after the periodic images have been undone.

For the order-$J_\pm^2$ and order-$J_\pm^3$ ice-manifold rows explicitly enumerated below, the possible transported dipoles are actually coarser than the microscopic quarter-lattice: their components lie in $\frac{1}{2}\mathbb{Z}$. In other words, the computed $\boldsymbol{\Delta} = 4\boldsymbol{\delta}$ components are always even. This is a property of the ice-preserving processes at these orders, not of a single nearest-neighbor hop. A single microscopic transverse hop still carries a quarter-lattice displacement, so the full microscopic source field is most naturally written with $4\boldsymbol{d}_{ij}$.

Lemma 2 (Gauge invariance of $\boldsymbol{\delta}$)**.** *The vector $\boldsymbol{\delta}_r$ is independent of the choice of home cell and of the gauge used to place the twist phases.*

Proof. Move a site coordinate by a cluster translation, $\boldsymbol{r}_i \mapsto \boldsymbol{r}_i + \boldsymbol{m}\boldsymbol{L}$ with $\boldsymbol{m} \in \mathbb{Z}^3$. If that site is raised, $\boldsymbol{\rho}$ changes by $+\boldsymbol{m}\boldsymbol{L}$; if it is lowered, $\boldsymbol{\rho}$ changes by $-\boldsymbol{m}\boldsymbol{L}$. The image bookkeeping of the virtual path changes by the opposite integer vector, because the same physical hop is now represented using a different periodic image. Therefore $\boldsymbol{N}\boldsymbol{L}$ changes by the negative of the change in $\boldsymbol{\rho}$, and the sum $\boldsymbol{\rho} + \boldsymbol{N}\boldsymbol{L}$ is unchanged. □

Proposition 2 (Transported-dipole character projection)**.** *For the clusters considered here, all coordinates lie on the quarter-lattice, so every transported dipole has components in $\frac{1}{4}\mathbb{Z}$. We therefore define the integer label*

$$\boxed{\boldsymbol{\Delta}_r \equiv 4\boldsymbol{\delta}_r \in \mathbb{Z}^3.} \tag{27}$$

This is only a change of units. For example, the winding four-loop in Fig. 11 has $\boldsymbol{\delta} = (0, 0, -\frac{1}{2})$, hence $\boldsymbol{\Delta} = (0, 0, -2)$. The condition $\boldsymbol{\Delta} = 0$ is exactly the same condition as $\boldsymbol{\delta} = 0$.

Couple an auxiliary character field $\boldsymbol{\theta}$ to this integer transported dipole. A perturbative row r acquires the phase

$$c_r(\boldsymbol{\theta}) = c_r e^{-i\boldsymbol{\theta}\cdot\boldsymbol{\Delta}_r}. \tag{28}$$

The operator-level average over $\boldsymbol{\theta} \in [0, 2\pi)^3$ keeps exactly the rows with $\boldsymbol{\delta}_r = 0$.

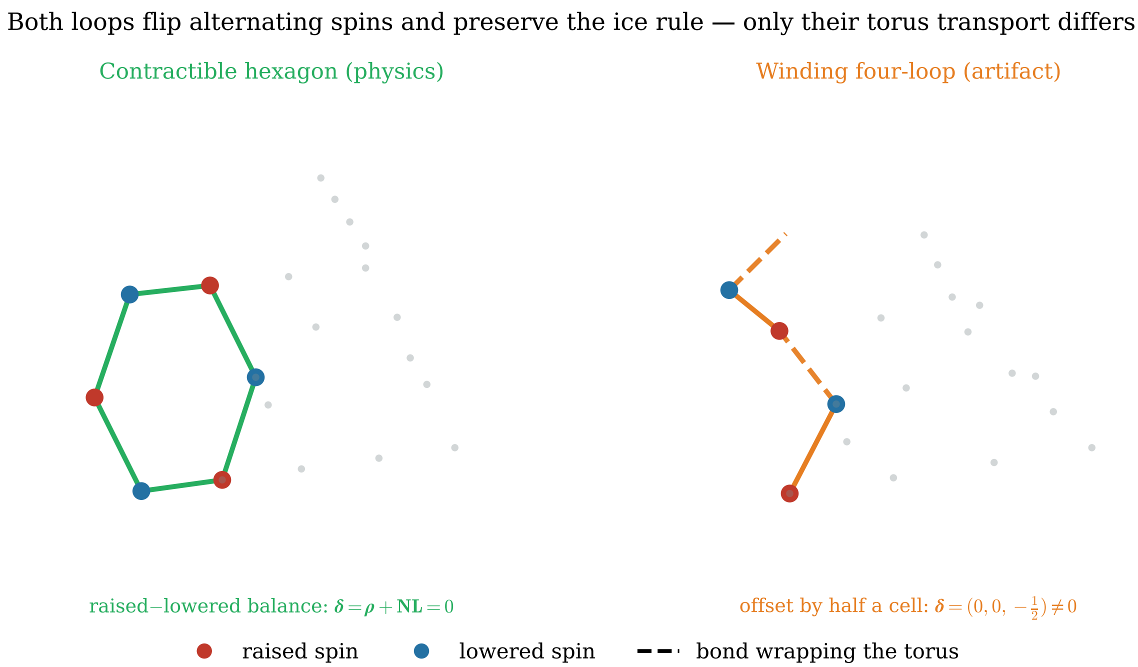

Figure 11: **The transported dipole tells the two loops apart, on the real 16-site cubic cluster.** Both loops flip alternating spins (red raised, blue lowered) and both return the system to the ice manifold. *Left:* the contractible hexagon is balanced—the raised and lowered spins average to the same point—so its transported dipole vanishes, $\boldsymbol{\delta} = \boldsymbol{\rho} + \boldsymbol{NL} = 0$. *Right:* the winding four-loop must cross the torus wall (dashed bonds); its raised and lowered spins are offset by half a unit cell, giving $\boldsymbol{\delta} = (0, 0, -\frac{1}{2}) \neq 0$. Selecting $\boldsymbol{\delta} = 0$ keeps the left process and discards the right one.

Proof. The characters of the abelian group $\mathbb{Z}^3$ are orthogonal:

$$\overline{c_r} = c_r \int_{[0,2\pi)^3} \frac{d^3\theta}{(2\pi)^3} e^{-i\boldsymbol{\theta}\cdot\boldsymbol{\Delta}_r} = c_r\, \delta_{\boldsymbol{\Delta}_r,0}.$$

Since $\boldsymbol{\Delta}_r = 4\boldsymbol{\delta}_r$, the condition $\boldsymbol{\Delta}_r = 0$ is equivalent to $\boldsymbol{\delta}_r = 0$. □

Remark 2 (How many discrete character points are needed?)**.** The exact integral over $\boldsymbol{\theta}$ can be replaced by a finite Fourier sum if the grid separates the transported-dipole sectors present in the calculation. For one component,

$$\frac{1}{M}\sum_{m=0}^{M-1} e^{-i(2\pi m/M)\Delta} = \begin{cases} 1, & \Delta = 0 \pmod{M}, \\ 0, & \Delta \neq 0 \pmod{M}. \end{cases}$$

Thus the finite grid projects onto $\Delta = 0$ only modulo M. In the order-$J_\pm^2$ and order-$J_\pm^3$ row data, the nonzero components of $4\boldsymbol{\delta}$ are ± 2. Therefore the $4\boldsymbol{\delta}$ convention needs a grid with $M \nmid 2$; the smallest choice is $M = 3$. If instead one worked only at the already-projected effective-row level and defined the integer label $\widetilde{\boldsymbol{\Delta}} = 2\boldsymbol{\delta}$, then the nonzero components would be ± 1 and an $M = 2$ character average would distinguish them from zero. We keep $4\boldsymbol{\delta}$ for the microscopic protocol because a single nearest-neighbor hop can carry a quarter-lattice displacement, so $4\boldsymbol{d}_{ij}$ is the natural integer bond label.

This is a pure aliasing issue. Consider one spurious four-loop component with $\delta_\mu = -\frac{1}{2}$. With the effective-row label $\widetilde{\Delta}_\mu = 2\delta_\mu = -1$, the two-point average gives

$$\frac{1}{2}\left(1 + e^{-i\pi(-1)}\right) = 0,$$

so the row is removed. With the microscopic integer label $\Delta_\mu = 4\delta_\mu = -2$, the same two-point average gives

$$\frac{1}{2}\left(1 + e^{-i\pi(-2)}\right) = 1,$$

so the row is not removed at all. The nonzero sector has aliased back into the zero sector modulo 2. The three-point average fixes this:

$$\frac{1}{3}\left(1 + e^{-i(2\pi/3)(-2)} + e^{-i(4\pi/3)(-2)}\right) = 0.$$

The perturbative row-table check on the 16-site cluster gives exactly this:

integer label	M	relative error to strict $\boldsymbol{\delta} = 0$
$4\boldsymbol{\delta}$	2	11.6476
$4\boldsymbol{\delta}$	3	1.1×10^{-15}
$2\boldsymbol{\delta}$	2	0
$2\boldsymbol{\delta}$	3	7.5×10^{-16}

Thus $M = 2$ was never universally correct; it was correct only after choosing the minimal effective-row charge $2\boldsymbol{\delta}$. Once the microscopic charge $4\boldsymbol{\delta}$ is used, $M = 2$ aliases the artifact and $M = 3$ is the smallest correct grid for the order-$J_\pm^2$ and order-$J_\pm^3$ row data.

Remark 3 (Why physical twists were insufficient)**.** A physical flat vector potential gives phases of the form $e^{-i\boldsymbol{A}\cdot\boldsymbol{\delta}_r}$. On the 16-site cluster the spurious four-loops have half-cell transported dipoles, so the corresponding exponent is half-integer in 2π units. Physical twist averaging therefore suppresses these rows but does not project them out exactly. The auxiliary character field in Eq. (28) is the finite-cluster source conjugate to the integer transported dipole $\boldsymbol{\Delta} = 4\boldsymbol{\delta}$.

Corollary 1 (Clean ice-manifold Hamiltonian)**.** *The artifact-free gauge Hamiltonian is*

$$\boxed{H_{\text{clean}}(J_\pm) = \sum_{r:\,\boldsymbol{\delta}_r=0} c_r J_\pm^{k_r} |t_r\rangle\langle s_r|.} \tag{29}$$

It keeps contractible hexagons and removes winding four-loops and wrapping hexagons.

11 Why the projection must act on the Hamiltonian

The specific heat is nonlinear in the spectrum, and the spectrum is a nonlinear function of the Hamiltonian. Therefore

$$C\left[T; \frac{1}{N_\phi}\sum_\phi H(\phi)\right] \neq \frac{1}{N_\phi}\sum_\phi C[T; H(\phi)] \tag{30}$$

in general. Averaging observables after diagonalization can leave positive even-in-coupling effects from a spurious matrix element. The projection must be applied at the operator level:

$$H_{\text{eff}} \longrightarrow H_{\text{clean}} \quad \text{first, then diagonalize.} \tag{31}$$

12 Practical algorithm

For a production ED calculation in the gauge sector:

1. Build the periodic pyrochlore cluster. Store each nearest-neighbor bond (i, j) and its image vector $\boldsymbol{n}_{ij}$.
2. Enumerate tetrahedra and the ice manifold $\mathcal{I}$.
3. Construct path-resolved Schrieffer–Wolff row tables $H_2, H_3, H_4, \ldots$ at $J_\pm = 1$.
4. For every row, compute $\boldsymbol{\rho}$, $\boldsymbol{N}$, and $\boldsymbol{\delta} = \boldsymbol{\rho} + \boldsymbol{N}\boldsymbol{L}$.
5. Assemble either the bare matrix

   $$H_{\text{eff}}^{\text{bare}}(J_\pm) = \sum_r c_r J_\pm^{k_r} |t_r\rangle\langle s_r|$$

   or the clean matrix in Eq. (29).
6. Diagonalize the ice-manifold matrix. The $2 \times 2 \times 2$ FCC cluster has only 2970 ice states, so dense diagonalization is exact and cheap.
7. Compute $C(T)$, entropy, and low-energy $S^{zz}(\boldsymbol{q}, \omega)$ from the clean eigenstates.

13 The campaign: a holonomy-resolved microscopic ED protocol

The effective-Hamiltonian calculation gives the target: the clean gauge sector is the contractible-hexagon manifold. The real ED campaign is to obtain the same object without first expanding the Hamiltonian by hand. The microscopic protocol should take the full Hamiltonian, diagonalize its low-energy ice-like band, and then remove precisely the components whose only reason for existing is torus winding. The organizing principle is not a numerical counterterm; it is the topology of a compact $U(1)$ gauge theory on a three-torus.

13.1 Gauge-theory interpretation of the ice manifold

The pyrochlore lattice is the line graph of the diamond lattice. A spin on a pyrochlore site is naturally a directed electric field on a diamond link $\ell = (rr')$:

$$E_\ell = S_i^z \tag{32}$$

up to a fixed convention for the link orientation. The ice rule is Gauss' law,

$$(\nabla \cdot E)_r = \sum_{\ell \ni r} \eta_{r\ell} E_\ell = 0. \tag{33}$$

Thus the ice manifold is the zero-charge sector of a compact lattice gauge theory. The transverse exchange is the microscopic quantum dynamics that, after virtual spinon fluctuations are integrated out, generates magnetic terms of the form

$$H_{\text{eff}}^{\text{bulk}} = -g_{\text{hex}} \sum_{p \in \text{hexagons}} \cos[(\nabla \times A)_p] + \cdots . \tag{34}$$

Here A_ℓ is the compact vector potential conjugate to E_ℓ. Equation (34) is local: each plaquette p is a contractible diamond hexagon. Local magnetic dynamics cannot know about a global torus cycle.

On a torus there are also harmonic, topological data. Define electric fluxes through the three noncontractible cuts,

$$W_\mu = \sum_{\ell \in \Sigma_\mu} E_\ell, \qquad \mu = x, y, z. \tag{35}$$

Contractible hexagon flips preserve $\boldsymbol{W} = (W_x, W_y, W_z)$ exactly. A loop that winds around the torus is different: it carries a nontrivial homology class in

$$H_1(T^3, \mathbb{Z}) = \mathbb{Z}^3. \tag{36}$$

Such a process is a Wilson line around a noncontractible cycle, not a local plaquette. In the thermodynamic gauge theory it is a finite-size tunnelling event between global sectors, and its amplitude should vanish exponentially with linear size. In a tiny periodic cluster, however, the same noncontractible path can be only four sites long, so it appears at the wrong power of $J_\pm$.

In plain language

The physical photon comes from small magnetic curls: contractible hexagons. The artifact comes from a Wilson loop that wraps around the whole torus but looks short because the torus is too small. The two processes can flip the same number of spins locally, but they live in different topology classes.

13.2 From flat holonomy to transported-dipole characters

Threading a physical twist through the periodic cluster is the same as coupling the spins to a flat background gauge field. A flat connection has zero local magnetic field but nontrivial holonomies

$$\Phi_\mu = \oint_{\gamma_\mu} \boldsymbol{A} \cdot d\boldsymbol{r} = \phi_\mu \tag{37}$$

around the three fundamental torus cycles. A contractible process sees no net holonomy. A process with winding vector $\boldsymbol{w} \in \mathbb{Z}^3$ picks up the character

$$\chi_{\boldsymbol{w}}(\boldsymbol{\phi}) = e^{-i\boldsymbol{w}\cdot\boldsymbol{\phi}}. \tag{38}$$

This is the correct language for integer homology. The finite-size loops found above are slightly subtler. On the 16-site cubic cluster their transported dipoles are half-cell displacements: in the integer coordinates

$$\boldsymbol{\Delta}_r \equiv 4\boldsymbol{\delta}_r \tag{39}$$

the offending rows have components $\Delta_\mu = 0, \pm 2$, whereas the clean hexagons have $\boldsymbol{\Delta} = 0$. An ordinary 2π-periodic flat holonomy is not orthogonal to a half-integer displacement. This is why physical twist averaging suppresses the artifact but does not reproduce the strict $\boldsymbol{\delta} = 0$ projector.

The factor 4 has no independent physical content. It is the denominator of the coordinate grid used for the pyrochlore sites. The exact projector needs a compact character variable with 2π periodicity:

$$\int_0^{2\pi} \frac{d\theta}{2\pi} e^{-i\theta n} = \delta_{n,0} \qquad (n \in \mathbb{Z}).$$

If we used the real displacement $\delta_\mu = -\frac{1}{2}$ directly, the exponent $e^{-i\theta\delta_\mu}$ would not be a single-valued character on the 2π-periodic θ circle. Multiplying by 4 converts all possible finite-cluster transported dipoles into integer labels. Since multiplication by 4 is invertible as a test of zero, it changes no physics in the projection: $\boldsymbol{\Delta} = 0$ if and only if $\boldsymbol{\delta} = 0$.

The exact character variable for the finite cluster is therefore not the integer winding $\boldsymbol{w}$, but the integer transported dipole $\boldsymbol{\Delta} = 4\boldsymbol{\delta}$:

$$\chi_{\boldsymbol{\Delta}}(\boldsymbol{\theta}) = e^{-i\boldsymbol{\theta}\cdot\boldsymbol{\Delta}}, \qquad \boldsymbol{\Delta} \in \mathbb{Z}^3. \tag{40}$$

Equivalently, the effective-Hamiltonian rows decompose into Fourier characters of $\boldsymbol{\Delta}$:

$$H_{\text{eff}}(\boldsymbol{\theta}) = \sum_{\boldsymbol{\Delta}\in\mathbb{Z}^3} H_{\text{eff}\,\boldsymbol{\Delta}}\, e^{-i\boldsymbol{\theta}\cdot\boldsymbol{\Delta}}. \tag{41}$$

The clean local gauge Hamiltonian is the trivial transported-dipole character:

$$H_{\text{eff}\,\text{local}} = H_{\text{eff}\,\boldsymbol{\Delta}=0} = \int_{[0,2\pi)^3} \frac{d^3\theta}{(2\pi)^3}\, H_{\text{eff}}(\boldsymbol{\theta}). \tag{42}$$

The important word is still *operator*. Equation (42) is not an average of spectra. It is a projection in the group algebra of transported-dipole characters.

Proposition 3 (Transported-dipole character projection)**.** *Assume the low-energy ice-band operator admits the decomposition Eq. (41). Averaging the operator over the character field* $\boldsymbol{\theta}$ *keeps exactly the zero-transport component and removes all nonzero transported dipoles:*

$$\int \frac{d^3\theta}{(2\pi)^3}\, H_{\text{eff}}(\boldsymbol{\theta}) = H_{\text{eff}\,\boldsymbol{\Delta}=0}.$$

Proof. The characters $e^{-i\boldsymbol{\Delta}\cdot\boldsymbol{\theta}}$ are orthonormal on the three-torus:

$$\int \frac{d^3\theta}{(2\pi)^3} e^{-i(\boldsymbol{\Delta}-\boldsymbol{\Delta}')\cdot\boldsymbol{\theta}} = \delta_{\boldsymbol{\Delta},\boldsymbol{\Delta}'}.$$

Applying this identity to Eq. (41) selects the coefficient with $\boldsymbol{\Delta} = 0$. Contractible hexagons have $\boldsymbol{\Delta} = 0$; wrapping finite-size loops have $\boldsymbol{\Delta} \neq 0$. □

13.3 Pulling the microscopic low band back to the ice basis

For the full microscopic Hamiltonian, the virtual paths are not explicitly enumerated. Instead we expose their transported dipole by diagonalizing a character-deformed Hamiltonian. For a nearest-neighbor bond represented by the minimum-image displacement

$$\boldsymbol{d}_{ij} = \boldsymbol{r}_j - \boldsymbol{r}_i - \boldsymbol{n}_{ij}\boldsymbol{L},$$

the microscopic move $S_i^+ S_j^-$ raises site i and lowers site j, so its transported-dipole increment is $-\boldsymbol{d}_{ij}$. The deformation whose virtual paths acquire the character $e^{-i\boldsymbol{\theta}\cdot\boldsymbol{\Delta}}$ is therefore

$$H_{\text{mic}}^{\Delta}(\boldsymbol{\theta}) = J_{zz}\sum_{\langle ij\rangle} S_i^z S_j^z - J_{\pm}\sum_{\langle ij\rangle}\left[e^{+i\boldsymbol{\theta}\cdot(4\boldsymbol{d}_{ij})}S_i^+ S_j^- + e^{-i\boldsymbol{\theta}\cdot(4\boldsymbol{d}_{ij})}S_i^- S_j^+\right], \tag{43}$$

where $4\boldsymbol{d}_{ij}$ is an integer vector on the clusters considered here. This is not an ordinary physical flux insertion. It is an auxiliary source field conjugate to the finite-cluster transported dipole. Physical flat twists are still useful diagnostics, but Eq. (43) is the deformation that reduces to the exact $\boldsymbol{\delta} = 0$ projection in perturbation theory.

Let $\{|\psi_n(\boldsymbol{\theta})\rangle\}_{n=1}^{N_{\mathcal{I}}}$ be the lowest ice-like band of $H_{\text{mic}}^{\Delta}(\boldsymbol{\theta})$, with energies $E_n(\boldsymbol{\theta})$. Let

$$P_{\mathcal{I}} = \sum_{\alpha\in\mathcal{I}} |\alpha\rangle\langle\alpha|$$

be the projector onto the fixed classical ice basis. Define the projected band matrix

$$X_{\alpha n}(\boldsymbol{\theta}) = \langle\alpha|\psi_n(\boldsymbol{\theta})\rangle. \tag{44}$$

If the microscopic low band is really the dressed ice manifold, then

$$G(\boldsymbol{\theta}) = X^{\dagger}(\boldsymbol{\theta})X(\boldsymbol{\theta}) \tag{45}$$

is nonsingular and close to the identity. The Loewdin-orthonormalized pullback is

$$Q(\boldsymbol{\theta}) = X(\boldsymbol{\theta})\, G^{-1/2}(\boldsymbol{\theta}), \qquad Q^{\dagger}Q = 1. \tag{46}$$

The full-Hamiltonian band operator, represented in one fixed ice basis, is

$$H_{\text{band}}(\boldsymbol{\theta}) = Q(\boldsymbol{\theta})\,\text{diag}\{E_n(\boldsymbol{\theta})\}\,Q^{\dagger}(\boldsymbol{\theta}). \tag{47}$$

This is the object to filter. It contains all orders in $J_{\pm}/J_{zz}$ that are present in the microscopic ED data, but it is expressed on the same ice manifold on which the gauge theory acts.

13.4 The clean microscopic operator

The proposed microscopic artifact-removal prescription is

$$\boxed{{H_{\text{band}}}^{\text{clean}} = \int_{[0,2\pi)^3} \frac{d^3\theta}{(2\pi)^3}\, H_{\text{band}}(\boldsymbol{\theta}).} \tag{48}$$

Equation (48) is the full-Hamiltonian analogue of Eq. (29). In the perturbative limit it reduces to the $\boldsymbol{\delta} = 0$ row projection. Away from the strict perturbative limit it is still meaningful: it extracts the component of the dressed ice band that is neutral under all transported-dipole characters.

For numerical work the integral is replaced by a Fourier quadrature,

$$H_{\text{band}}{}_M^{(0)} = \frac{1}{M^3} \sum_{m_x,m_y,m_z=0}^{M-1} H_{\text{band}}\left(\frac{2\pi m_x}{M}, \frac{2\pi m_y}{M}, \frac{2\pi m_z}{M}\right), \qquad H_{\text{band}} = H_{\text{band}}(\boldsymbol{\theta}). \tag{49}$$

For the 16-site campaign below we use the minimal effective-row label

$$\widetilde{\boldsymbol{\Delta}} = 2\boldsymbol{\delta}.$$

The spurious four-loop sectors then carry $\widetilde{\Delta}_\mu = \pm 1$, so the two-point grid $M = 2$ distinguishes them from zero. This is the smallest character average that exactly reproduces the strict $\boldsymbol{\delta} = 0$ row projector through order $J_\pm^3$. With the alternative microscopic integer label $\boldsymbol{\Delta} = 4\boldsymbol{\delta}$, the same artifacts carry $\Delta_\mu = \pm 2$, so $M = 2$ aliases them back to zero and one needs $M = 3$.

In larger clusters or at higher perturbative order, M should be increased until $C(T)$, the low-band spectrum, and selected matrix elements of $H_{\text{band}}{}_M^{(0)}$ converge.

13.5 Concrete ED protocol

The full microscopic calculation should therefore be organized as follows.

1. Choose a periodic cluster and construct a fixed ice basis $\{|\alpha\rangle\}$.

2. For each character field $\boldsymbol{\theta}$ on the grid, build the source-deformed microscopic Hamiltonian in the full fixed-$S^z_{\text{tot}} = 0$ Hilbert space. In the adopted $2\boldsymbol{\delta}$ protocol, replace $4\boldsymbol{d}_{ij}$ by $2\boldsymbol{d}_{ij}$ in Eq. (43) and use the $M = 2$ grid $\theta_\mu \in \{0, \pi\}$.

3. Diagonalize the lowest $N_{\mathcal{I}}$ states. Record the spectral separation between the $N_{\mathcal{I}}$th and $(N_{\mathcal{I}} + 1)$st levels whenever possible.

4. Project the eigenvectors onto the ice basis and monitor

$$\lambda_{\min}\left[X^\dagger(\boldsymbol{\theta})X(\boldsymbol{\theta})\right].$$

   If this value is too small, the chosen coupling is outside the regime where a clean ice-band pullback is controlled.

5. Construct $H_{\text{band}}(\boldsymbol{\theta})$ using Eq. (47).

6. Perform the operator-level character projection Eq. (49). Diagonalize only after this step.

7. Validate the method by the weak-coupling limit:

$$H_{\text{band}}{}_M^{(0)} - \text{const.} = H_{\text{clean}} + O(J_\pm^4/J_{zz}^3)$$

   when M resolves the winding characters present through third order.

8. Use the resulting eigenstates for the low-temperature gauge-sector thermodynamics and for observables such as $S^{zz}(\boldsymbol{q}, \omega)$ that do not leave the ice manifold.

13.6 First microscopic test on the 16-site cluster

We performed this test with the local QED package on the 16-site cubic cluster at $J_\pm = -0.05 J_{zz}$. The full fixed-$S^z_{\mathrm{tot}} = 0$ Hilbert space has dimension

$$\binom{16}{8} = 12870,$$

and the ice band has $N_{\mathcal{I}} = 90$ states. Dense partial diagonalization was used to obtain the lowest 90 eigenvectors at each source-field point.

The perturbative row-table test first establishes the correct source field. On this cluster, physical flat twists and integer winding projectors do not match the strict $\boldsymbol{\delta} = 0$ Hamiltonian. The transported-dipole character does:

projector	T_{peak}/J_{zz}	rel. error to $\boldsymbol{\delta} = 0$	comment
all rows	0.0100564	11.6476	winding four-loop present
integer $N = 0$	0.0046462	5.8238	boundary winding only
physical $\boldsymbol{A}\cdot\boldsymbol{\delta}$, $M = 6$	0.0016563	3.0740	suppresses, not exact
$2\boldsymbol{\delta}$ character, $M = 2$	0.0016045	0	minimal row-level charge
$\boldsymbol{\Delta} = 4\boldsymbol{\delta}$ character, $M = 3$	0.0016045	0	exact through order 3

We tested both microscopic source conventions. The adopted $2\boldsymbol{\delta}$ run used Eq. (43) with $4\boldsymbol{d}_{ij}$ replaced by $2\boldsymbol{d}_{ij}$ on the $M = 2$ grid, i.e. 8 source-field points. The $\boldsymbol{\Delta} = 4\boldsymbol{\delta}$ comparison run used Eq. (43) on the $M = 3$ grid, i.e. 27 source-field points. The low-band pullback was well conditioned in both cases; for the $2\boldsymbol{\delta}$ run,

$$\min_{\boldsymbol{\theta}} \lambda_{\min}\left[X^\dagger(\boldsymbol{\theta}) X(\boldsymbol{\theta})\right] = 0.8872896692,$$

and the QED eigenvectors were orthonormal to 2.3×10^{-14}. The resulting specific-heat peaks are

operator	T_{peak}/J_{zz}	comment
full QED band, $\boldsymbol{\theta} = 0$	0.0129476	matches all-row physics
full QED band, $2\boldsymbol{\delta}$ character $M = 2$	0.0012227	near clean scale, not identical
full QED band, $4\boldsymbol{\delta}$ character $M = 3$	0.0012222	better operator match
perturbative all rows	0.0100564	four-loop artifact
perturbative $\boldsymbol{\delta} = 0$	0.0016045	clean hexagon target

The equality claim in the perturbative table is an operator statement inside the order-$J_\pm^2$ and order-$J_\pm^3$ ice-manifold row table. It does not imply that the finite-$J_\pm$ full-Hamiltonian specific heat peak must land exactly at the perturbative $\boldsymbol{\delta} = 0$ peak. The full QED low band contains Schrieffer-Wolff dressing and higher-order terms beyond the truncated row table, and the specific heat is a nonlinear function of the final projected spectrum. The appropriate full-Hamiltonian check is therefore whether the character projection removes the four-loop-scale peak and moves the result to the clean hexagon scale, not whether the peak positions are numerically identical at $J_\pm = -0.05 J_{zz}$.

At the operator level, after removing constants, the adopted $2\boldsymbol{\delta}$, $M = 2$ projected full QED band has relative Frobenius error 0.3527 against the perturbative $\boldsymbol{\delta} = 0$ matrix. Allowing one overall scale factor, the residual falls to 0.2612; the corresponding spectrum residual is also 0.2612. The $4\boldsymbol{\delta}$, $M = 3$ comparison run gives a slightly better operator residual, 0.1450 after scale fitting, but it requires 27 full ED points instead of 8. Thus the working protocol for the present 16-site

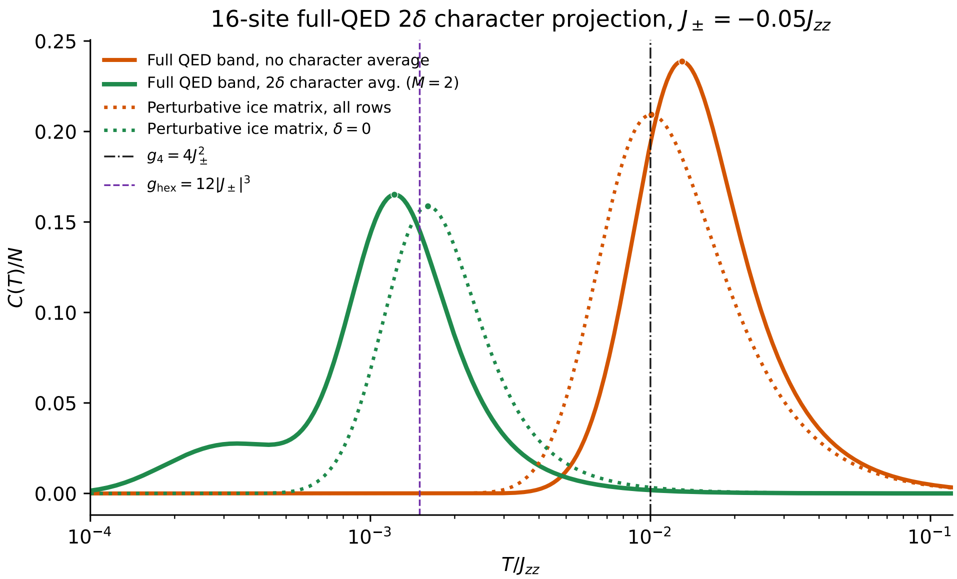

Figure 12: **Specific heat from the adopted $2\boldsymbol{\delta}$, $M = 2$ full-QED protocol.** The orange solid curve is the untwisted full microscopic low-band pullback. The green solid curve is the operator-level $2\boldsymbol{\delta}$ character average over the $M = 2$ grid. Dotted curves are the perturbative ice-manifold matrices with all rows and with the strict $\boldsymbol{\delta} = 0$ projector. The character average moves the low-temperature peak from the spurious four-loop scale toward the clean hexagon scale.

campaign is the cheaper and perturbatively minimal $2\boldsymbol{\delta}$, $M = 2$ character average. Both character protocols are a large improvement over the ordinary physical $M = 2$ holonomy average, whose operator error against $\boldsymbol{\delta} = 0$ was 5.4308.

Thus the first microscopic test supports the corrected campaign: full ED dresses the ice manifold, while the transported-dipole character projection removes the dominant finite-size four-loop and recovers the clean hexagon scale up to higher-order dressing of the microscopic band.